

OpenClaw AI Rock Paper Scissors Game Tutorial

Before starting, please confirm that CSI camera, OLED, RGB, and dialogue channel are already configured. This article only expands on the Jetson-side AI rock paper scissors game application.

Functions in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

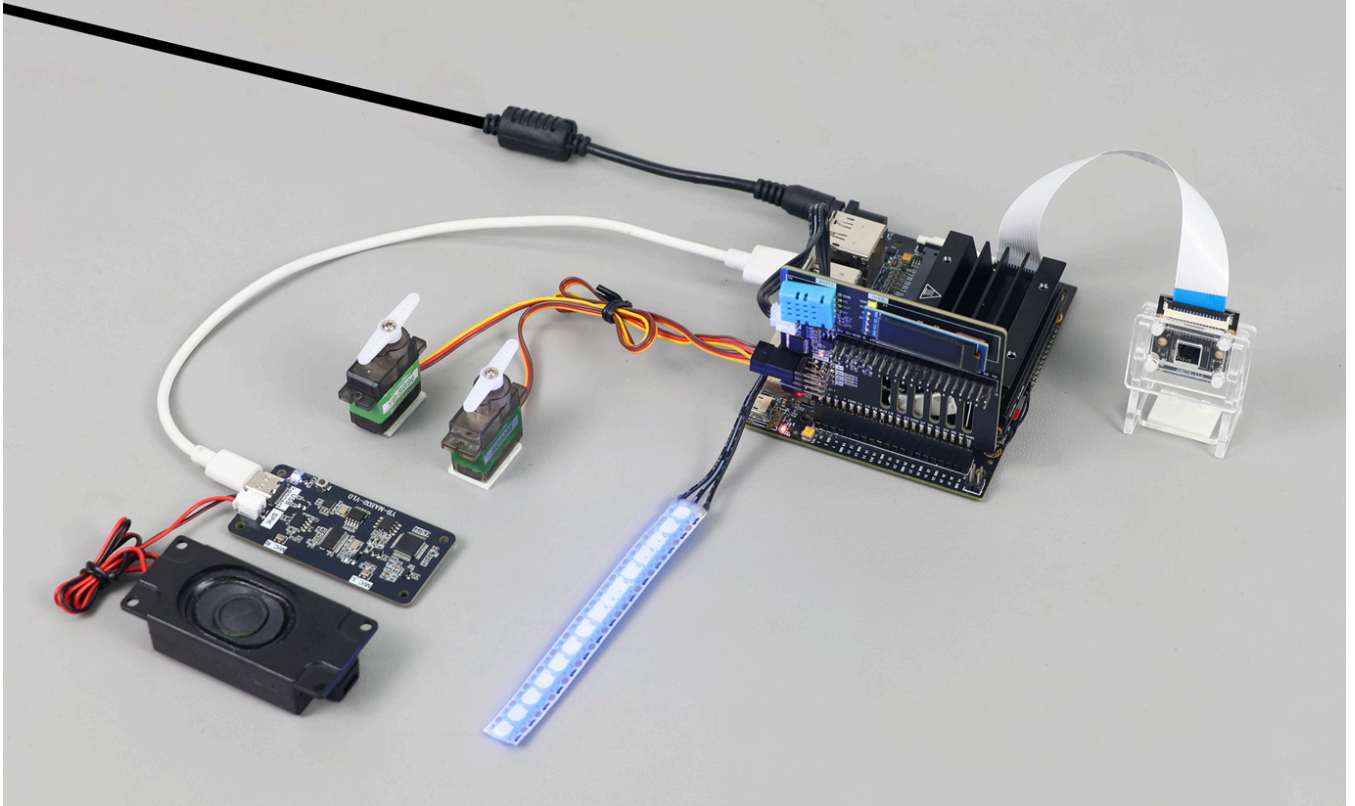
1. Tutorial Objectives

This article demonstrates the interactive gaming capability of OpenClaw on Jetson through "taking photos to recognize gestures + AI random moves + win/lose judgment + peripheral feedback". After completion, you can achieve:

- Take photos to recognize the rock, paper, or scissors the player played.
- AI makes random moves.
- Determine who won this round.
- OLED displays the round result.
- Feedback win/lose status through RGB.

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- CSI camera
- RGB light strip
- Optional: Servo
- Optional: AI voice interaction module + speaker
- Wiring diagram



3. Software Entry

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

Common commands:

```
/usr/local/bin/csi_snap snap  
camera tool action="describe"  
/usr/local/bin/oled text --text "You win"  
/usr/local/bin/rgb --format json color --color green  
/usr/local/bin/rgb --format json color --color red  
/usr/local/bin/rgb --format json color --color blue
```

Suggested feedback rules:

- AI wins: RGB lights up green.
- AI loses: RGB lights up red.
- Draw: RGB lights up blue.
- OLED displays AI win, You win, or Draw.

4. Basic Capability Test

Enter TUI:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔌 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
█
```

Test first:

- Take a photo
- OLED displays "Ready"
- RGB lights up blue

5. Three Dialogue Methods

Method	Suitable Scenario	Suggested Positioning
TUI	Local debugging game logic	Preferentially verify recognition and win/lose judgment
WhatsApp	Mobile remote interactive demonstration	Suitable for displaying images and text results
Voice	On-site interactive gaming	Suitable for directly saying "let's play rock paper scissors"

5.1 WhatsApp Method

You can send:

- Please play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture. Then judge whether you win or I win OLED displays who this round

20:27



< 2

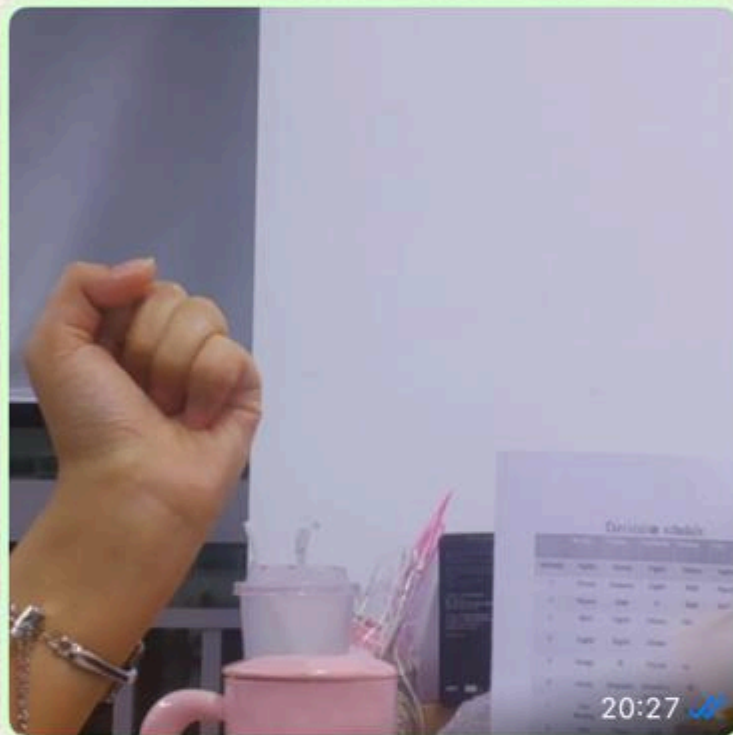


+86 153 3885 7526 (你)

给自己发消息

Please play rock paper scissors.
Take a photo to recognize my
gesture. You can randomly make a
gesture. Then judge whether you win
or I win OLED displays who this round.

20:27 ✓✓



I can see the user's hand gesture in
the image - it's a **fist**, which
represents **ROCK** in rock paper
scissors.

Now I need to:

1. Make a random gesture choice
(rock, paper, or scissors)
2. Determine the winner
3. Display the result on OLED





5.2 TUI Method

Run:

```
openclaw tui

root@jetson-yahboom:~# openclaw tui
OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

Recommended input:

- Let's play rock paper scissors
- You randomly make a move, take a photo to recognize my gesture, then determine who wins
- Rock paper scissors game, finally display the result on OLED



5.3 Voice Method

Execute the following command with the gateway running:

```
cd /root/yahboom_ws/src/largemodel  
python3 main.py
```

You can say:

- Let's play rock paper scissors
- See what I played, determine who won the rock paper scissors



6. Comprehensive Cases

6.1 Gesture Recognition

Experiment Objective: Recognize the player's current gesture.

Example Commands:

- Take a photo to recognize my gesture
- See if I played rock, paper, or scissors
- I've made my move, help me recognize it

6.2 AI Move

Experiment Objective: Let OpenClaw randomly generate the AI's move result.

Example Commands:

- You randomly make a gesture
- Let's play rock paper scissors, you make a move too
- You play one of rock, paper, or scissors

6.3 Win/Lose Judgment

Experiment Objective: Determine this round's result based on player gesture and AI gesture.

Example Commands:

- Let's play rock paper scissors
- Take a photo to recognize my gesture, you randomly play, then determine who wins

6.4 Peripheral Feedback

Experiment Objective: Display results using OLED and RGB.

Example Commands:

- Display rock paper scissors result on OLED
- If you win RGB lights up green, if you lose RGB lights up red
- If it's a draw RGB lights up blue

7. Common Problems

- Gesture recognition unstable: Keep the gesture fully in the frame, keep the background as simple as possible.
- Reply but no photo taken: Check if the `camera` tool was called, or if `/usr/local/bin/csi_snap` was used in the script.
- Only returns one sentence without peripheral feedback: Check if OLED and RGB actions were generated.